Region-Relative Verification of TinyTaxiNet Using Marabou 2.0

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Abstract

In formal verification of image-based neural networks, a SAT/UNSAT verdict is exact only for the input region encoded for the solver, and constructing that region is a consequential modeling choice. This paper treats input-region selection as an explicit experimental variable. We introduce a reusable Marabou workflow that reruns a fixed property family across six input abstractions, tags each verdict with the region that produced it, and audits recovered counterexamples against dataset-support signals. Evaluated on TinyTaxiNet, an open-source runway centerline-tracking surrogate, the workflow separates three verdict patterns. First, a too-loose region admits pixel combinations outside the released images: one heading-error violation disappears once the mean-centering constraint satisfied by every released image, mean(x)=0.5, is enforced. Second, reducing the input dimension accounts for several changes in the verification results. For example, two centered cross-track output-bound queries become UNSAT both inside an 8D autoencoder decoder-box and inside matched random 8D affine regions. Likewise, the local-robustness verification rate rises from 186/893 anchors to 847/893 under 8D local PCA, but is matched or exceeded by random subspaces of the same dimension. A dimension sweep shows that data-aligned directions begin to matter beyond $d \approx 16$, where PCA regions recover cross-track counterexamples that random subspaces at the same scale mostly miss. Third, a wrong-sign cross-track error survives every tested region tightening and is observed directly on images in the released simulator dataset. Instantiations on MNIST, VNN-COMP `mnist_fc`, and LARD V2 runway imagery reproduce the dimensionality pattern under matched random controls. The core implementation is released as a reusable module.

Keywords Neural network verification · Marabou · Image-based perception · Input-region tagging · Witness audit · Completeness-realism trade-off · TinyTaxiNet · MNIST

1 Introduction

In formal verification of image-based neural networks, solver verdicts are meaningful only with respect to the input region presented to the verifier. For piecewise-linear image networks, per-pixel boxes and L_∞ balls are straightforward to express in Marabou because they become lower and upper bounds on input variables, but they can contain pixel combinations that are far outside the image structure induced by a camera and preprocessing pipeline. Learned, local, or analytically constrained regions move in the opposite direction: they reduce loose pixel combinations seen by the solver, but their guarantees hold only over the fitted or encoded support model. Prior work exposes this issue through runway image-perturbation studies, generator-composed verification, analytic observation models, and surveys of vision-based autonomous-system verification [4,7,27,2,3]. This paper treats input-region construction as an explicit verification variable.

We present a region-tagged verification workflow for Marabou (Fig. 1). Each SAT/UNSAT verdict is reported with the input region that produced it, the same property family is rerun under multiple region tags, and recoverable SAT witnesses are audited using classifier probability, nearest-real-image L_2 distance, and per-pair Lipschitz violation fraction. A separate closed-form input-Jacobian and saturating-perturbation diagnostic accounts for postcondition effects, so output-axis asymmetries are not mistaken for architectural robustness.

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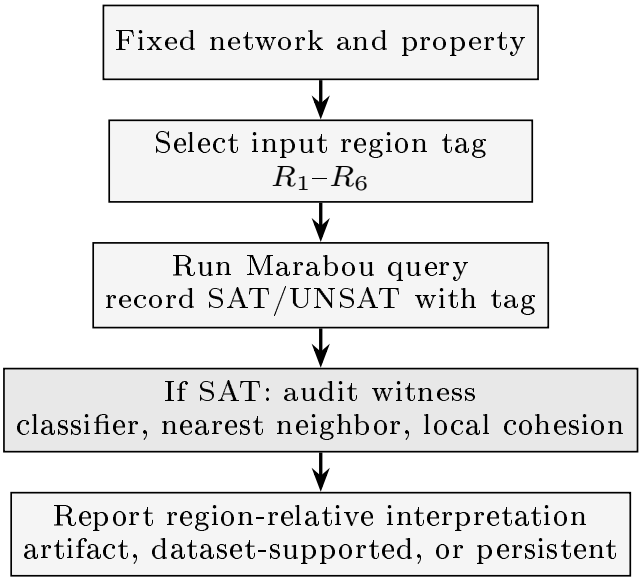


Fig. 1 Region-tagged verification workflow. The method holds the network and property fixed, varies the encoded input region, records every Marabou verdict with its region tag, and audits recovered SAT witnesses before interpreting the result.

The `region_audit.py` module implements this workflow for flattened image-input ReLU networks. To use it with a new model, the user supplies the network reader, data loader, subset rules, neighbor rule, and calibration data. The module uses these inputs to attach the selected region and run the specified query. It then records the verdict with its region tag and parameters and scores recovered counterexamples against a same-domain baseline.

The audit results help interpret counterexamples in relation to the simulator dataset. A counterexample recovered under a broad input region may depend on pixel combinations absent from the released images. If the failure persists under tighter regions and also appears in those images, it has direct empirical support.

We use TinyTaxiNet as the primary case study, an open-source surrogate for runway-perception tasks that has appeared in several TaxiNet-family verification studies. It maps 8×16 grayscale runway images to cross-track and heading error. Prior studies on TinyTaxiNet and related TaxiNet models adopt input abstractions tailored to their verification objectives, including per-image perturbation regions, activation-pattern-conditioned boxes, and generative front-end regions [5,6,7]. This paper instead holds the perception network and property family fixed while varying the encoded input region and auditing recovered witnesses.

This comparison separates a preprocessing artifact from an 8D decoder-box result in P1. In P3, reducing the query to a low-dimensional region raises the verification rate by more than a factor of four. The matched random-subspace control links this increase to dimensionality, while local PCA exposes more cross-track counterexamples than random 8D regions at the same scale. In P4, the wrong-sign cross-track mode survives every tested region tightening and appears in the simulator dataset.

The study addresses three questions.

- **RQ1: Verdict sensitivity.** How do Marabou SAT/UNSAT verdicts change when the same network and related tracking properties are evaluated under different input-region constructions?
- **RQ2: Counterexample support.** Which SAT witnesses are supported by empirical behavior in the released simulator dataset, and which depend primarily on loose pixel-space over-approximation?
- **RQ3: Audit value.** Which diagnostics are needed to interpret SAT witnesses produced under learned or structured regions?

To address these questions, we developed a verification protocol and reusable artifact using Marabou, with the following contributions:

1. **Region-tagged comparative protocol.** We define six region tags ranging from raw pixel boxes to latent, local, and linear-constraint regions, then verify the same property family under multiple regions. Each Marabou SAT/UNSAT verdict is paired with the input abstraction that produced it (Section 3.1), isolating changes caused by preprocessing invariants, learned support models, and structural filters.
2. **Postcondition-headroom diagnostic.** We give a closed-form input-Jacobian and saturating-perturbation diagnostic that separates input-region effects from specification-margin effects (Section 5.2.1).
3. **Witness-audit protocol.** We pair recovered SAT witnesses with classifier probability, nearest-real-image L_2 distance, and per-pair Lipschitz violation fraction (Sections 6.1, 6.3). The audit distinguishes regional witnesses that remain far from the simulator image distribution from witnesses with empirical dataset support.

We also test the workflow on a small MNIST ReLU MLP, comparing broad pixel regions with autoencoder and local-PCA regions (Section 5.6). The VNN-COMP `mnist_fc` experiment adds a check on an externally supplied image classifier (Section 5.7). For runway imagery, we use LARD V2 to examine broad-region counterexamples and audit their support in a dataset independent of VerifyGAN (Section 5.8). Section 8 describes the released code, results, and tests, along with the module interface and runtime details.

Evidence block	Role	Claim scope
TinyTaxiNet	Main P1/P3/P4 case study	Region-relative verdict interpretation
MNIST MLP	Second-model protocol check	Workflow transfer; domain-specific audit thresholds
VNN-COMP `mnist_fc`	External image classifier	Bounded low-dimensional removal on a non-authored network
LARD V2	External runway images	Portability and audit check on a pitch-attitude target
Controlled oracle	Known-truth calibration	Separates true injected failures from artifacts
Reusable module	Region and audit implementation	Reusable protocol for flattened image-input ReLU networks

Table 1 Evidence map and claim scope for the paper. The external checks instantiate the protocol across data sources and models; claims remain per-instantiation.

The rest of the paper is organized as follows. Section 2 describes TinyTaxiNet and the released preprocessing pipeline. Sections 3 and 4 define the region tags and formal properties. Section 5 gives the main TinyTaxiNet probes and the MNIST, `mnist_fc`, and LARD instantiations. Section 6 audits witnesses and stress-tests learned and math-encoded regions. Section 7 compares the protocol with related verification approaches, Sections 8–9 describe reproducibility and calibration checks, Section 10 discusses validity scope and future work, and Section 11 concludes.

2 System Under Verification and Preprocessing

TinyTaxiNet is an open-source feedforward ReLU network distributed in the sisl/VerifyGAN repository [7] as a compact surrogate for the Boeing TaxiNet perception network used in autonomous center-line tracking [6,8]. We use the repository's TinyTaxiNet artifacts, not the GAN-based verification method of Katz et al. [7]. The network implements

$$f : \mathbb{R}^{128} \to \mathbb{R}^2$$

by mapping a flattened 8×16 grayscale runway image through three hidden layers (16, 8, 8 neurons with ReLU activation) to two outputs:

- y_0: cross-track error (CTE, meters from centerline)
- y_1: heading error (HE, degrees off alignment)

The network has 32 ReLU neurons ($16 + 8 + 8$) and 194 total variables in Marabou's encoding (inputs 0–127, outputs 192–193). Verification operates on the released preprocessed 128-vectors described below.

2.1 Data and Preprocessing

We use the TinyTaxiNet weights and publicly available dataset released by the SISL team in the sisl/VerifyGAN repository [7]. This dataset contains 10,000 simulator-generated training images with CTE, HE, and DTP labels.

Preprocessing. Each raw X-Plane screenshot is cropped, resized, converted to grayscale, downsampled to 8×16, then per-image mean-centered and clipped:

$$x \leftarrow \mathrm{clip}(x - \overline{x} + 0.5,\ 0,\ 1) \tag{1}$$

The preprocessed image is flattened into a 128-element vector, stored in HDF5, and loaded for verification.

Dataset and preprocessing checks. All 10,000 images have pixel mean $= 0.5000$, confirming the mean-centering invariant. The `.nnet` file embeds no additional normalization, and we set `normalize=False` in Marabou to match the released inputs.

3 The Region Spectrum

Marabou and other complete neural-network verifiers [10,11,12] answer queries of the form $\forall x \in R.P(f(x))$, where f is the network and R is the encoded input region. Marabou returns SAT when a concrete counterexample exists in R and UNSAT when the property holds for all inputs in R. The six tags below identify the input region R used in each query.

3.1 Verification regions

Formal definitions, Marabou encodings, and calibration details are in Appendix A.

- R_1 **(raw per-pixel box):** independent per-pixel bounds, either a subset envelope $R_1(S)$ or a per-image ball $R_1(x_0, \varepsilon)$.
- R_2 **(R_1 + mean invariant):** R_1 plus the preprocessing equality mean(x)=0.5.
- R_3 **(AE decoder-box, mean-constrained):** subset-specific ReLU autoencoder decoder composed with TinyTaxiNet over the train-only latent box.

- R_4 **(local-PCA per-image, mean-constrained):** top-d=8 same-state-neighbor principal directions around the query image, scaled by $\alpha\sigma$.
- R_5 **(label-conditioned box):** per-pixel [min, max] over same-state neighbors of the anchor, intersected with clipping and the mean invariant.
- R_6 **(math-encoded, mean-constrained):** box-plus-mean intersected with linear-in-pixel image-structure filters: TV, Fourier high-frequency L_1, photometric residual, per-row mean, per-cell mean, or per-pair Lipschitz bounds.

Tags are parameterized by their calibration data, a subset S, an anchor (x_0, ε), or both. Dataset-only observations are reported as empirical baselines, not solver regions, because they invoke no Marabou query and produce no SAT/UNSAT verdict.

3.2 Dataset query domains

Table 2 lists the dataset-level domains used to calibrate region bounds, latent boxes, PCA bases, and empirical filter thresholds.

Subset	Definition	Used by	Images
C_{centered}	$\lvert\text{CTE}\rvert < 1.0\,\text{m}$	P1, P6	893
C_{left}	$\text{CTE} > 2.0\,\text{m}$	P4	4,128
C_{right}	$\text{CTE} < -2.0\,\text{m}$	P4	4,099
$C_{\text{off-center}}$	$\lvert\text{CTE}\rvert > 1.0\,\text{m}$	R_3 ($\text{AE}_{\text{off-center}}$)	9,107
L_∞ ball	$\lVert x - x_0\rVert_\infty \le \varepsilon$	P2, P3	per image

Table 2 Input subsets and query domains used to calibrate the region tags. Subset rows define dataset-level envelopes; the L_∞ row is per-image.

3.3 Perturbation model and threshold rationale

Throughout this paper, ε denotes an L_∞ radius in the preprocessed network-input space. We use these radii to assess the network's sensitivity around real preprocessed images; they do not measure the magnitude of raw-camera or weather perturbations. P2 and P3 use $\varepsilon \in \{0.02, 0.05, 0.10\}$. We compare these radii with L_∞ distances between same-state neighbors in the centered subset. For each image, the neighbor is the closest other image satisfying $|\Delta\text{CTE}| \le 0.5\,\text{m}$ and $|\Delta\text{HE}| \le 5°$. The resulting distances have 10th, 25th, 50th, and 90th percentiles of 0.033, 0.060, 0.103, and 0.189, respectively. By comparison, the median L_∞ distance between random image pairs is 0.271. The chosen radii therefore test perturbations ranging from below the 10th percentile to approximately the median same-state-neighbor distance. All three remain well below the median distance between random image pairs.

P1 and P3 test whether predictions stay within broad tracking limits: $|\text{CTE}| < 10\,\text{m}$ and $|\text{HE}| < 90°$. The 10 m limit corresponds to about 22% of the approximately 46 m width of Grant County Runway 04, while 90° represents complete loss of alignment. P2 tests local correctness against the reference label with tighter tolerances of $|\Delta\text{CTE}| < 1.0\,\text{m}$ and $|\Delta\text{HE}| < 5.0°$. These output thresholds follow Kadron et al. [6] where applicable. The remaining thresholds are case-study choices for TinyTaxiNet: P4 left/right at $|\text{CTE}| > 2.0\,\text{m}$, P6 binary-search tolerance 0.01.

4 Formal Specifications

The baseline property set contains five Marabou encodings. Table 3 lists the exported baseline queries by property class.

- **Invariance / output-bound properties** (P1, P3): universal claims that for all inputs in a region, the output lies within an absolute box. Negation is a disjunction over bound violations.
- **Sign and relational properties** (P2, P4): universal claims about a relation between input/output or between output components. P2 relates output to a reference label; P4 relates output sign to input-region label.
- **Reachability / output-range property** (P6): an existential query about what outputs are reachable. P6 sweeps thresholds to find the tight reachable range for each output axis.

Property	Encoded query	Queries
P1-global	$x \in C_{\text{centered}}$; prove $\lvert y_0\rvert < 10$ and $\lvert y_1\rvert < 90$ with one sub-query per violated side	4
P1-patterned	P1 over each observed centered 32-bit activation pattern	96
P2	$\lVert x - x_0\rVert_\infty \le 0.02$; prove $\lvert y_0 - y_0^*\rvert < 1.0$ and $\lvert y_1 - y_1^*\rvert < 5.0$	160
P3	$\lVert x - x_0\rVert_\infty \le \varepsilon$; prove the P1 output envelope around sampled anchors	160
P4	$x \in C_{\text{left/right}}$; prove the CTE output sign matches the offset side	2
P6	$x \in C_{\text{centered}}$; binary-search tight reachable intervals for y_0 and y_1	52

Table 3 Baseline exported formal queries by property class. Strict inequalities are negated using Marabou's non-strict bound API. P2/P3 use 20 evenly spaced baseline images; P6 uses binary search with tolerance 0.01.

Property use. P1, P3, and P4 are evaluated under multiple region constructions. P1-patterned partitions C_{centered} by the 32-bit activation pattern across all

hidden layers and verifies each observed pattern; only the 24 centered-image patterns are visited, the others have no empirical evidence of occurrence. Property numbering follows the released artifact identifiers; P5 was superseded during revision and is not part of the baseline set.

5 Three Probes

The primary results are P1, P3, and P4 (Table 4). P1 separates a preprocessing artifact from an 8D decoder-box claim, P3 measures sensitivity to region dimension and orientation, and P4 tracks a failure mode that survives every tested region tightening. Section 5.6 applies the same protocol to MNIST as a transfer check.

5.1 P1: box artifact and AE-region result

P1-global evaluates whether every input in the centered per-pixel box remains inside the Kadron-style tracking envelope. Over the raw region R_1, the property is **violated**: three of four sub-queries return SAT.

- P1a (CTE ≥ 10): SAT. Witness: CTE = 10.0, HE = $-52.5°$.
- P1b (CTE ≤ -10): SAT. Witness: CTE = -10.0, HE = $15.0°$.
- P1c (HE $\geq$ 90): **UNSAT**. No centered input produces HE $\geq 90°$.
- P1d (HE ≤ -90): SAT. Witness: CTE = 18.5, HE = $-90.0°$.

P1-patterned tests whether separating the centered region by activation pattern is enough to establish the P1 bounds. The partition uses the 24 observed Prophecy/Kadron-style patterns [13,6], each defined by the complete 32-bit sign vector across the hidden ReLUs. The solver verifies **16 of 24 patterns**, which cover 378 of 893 centered images. The remaining 8 patterns fail on CTE, and the four largest failed patterns contain about 86% of the images in violated regions. The verified patterns therefore cover a minority of the centered dataset; most images belong to patterns whose regions still admit CTE-bound violations.

Mean-constrained follow-up. Enforcing mean(x) = 0.5 changes P1d from SAT under R_1 to UNSAT under R_2 and R_3. The HE $\leq -90°$ counterexample therefore depends on inputs excluded by this preprocessing invariant. Across the 10,000 released images, the maximum absolute heading error is $62.6°$, below $90°$ (Section 6.1).

P1a and P1b remain SAT under R_1 and R_2 and become UNSAT under the 8D AE decoder-box region R_3. The centered empirical CTE-error maximum is 4.64 m, below the 10 m threshold; the 23 full-dataset images exceeding 10 m are all off-center. Thus, the lateral SAT witnesses are not failures directly observed on centered dataset images.

To test whether the decoder is responsible for the verdict change, we use matched random 8D affine regions. These regions have deterministic zero-mean orthonormal bases and are calibrated to the centered subset's projection box as $x = \mu + V^\top z$ with $0 \leq x \leq 1$. Both queries become UNSAT across three random seeds and under the corresponding PCA 8D affine region. The regions still permit substantial output variation: a reproducible corner/random search reaches CTE magnitudes of 7.9 m for PCA and 2.4–3.9 m for random, while Marabou proves that the 10 m bound is unreachable in each region. The 8D restriction therefore accounts for the P1a/P1b verdict change without requiring a learned decoder. PCA reaches closer to the bound than random, motivating the dimension and direction comparisons in P3.

5.2 P3: region-dependent robustness

P3 asks whether predictions remain inside the broad ± 10 m, $\pm 90°$ output envelope under local perturbation. We evaluate matched centered anchors under unconstrained L_∞ balls, mean-constrained and clipped L_∞ balls, and local-PCA regions calibrated to same-state image variation. The full-centered mean+clip run verifies 892/893 images at $\varepsilon = 0.02$, 186/893 at $\varepsilon = 0.05$, and 0/893 at $\varepsilon = 0.10$.

The difference between P2 and P3 reflects their output tolerances: P2 tests a 1 m correctness band, while P3 tests a 10 m output envelope. On the 20-image sample, P3 verifies 3–4/20 at $\varepsilon = 0.05$, compared with 0/20 for P2 at $\varepsilon = 0.02$. All P3 failures under these L_∞ balls are CTE-driven. Under the mean-constrained ball, 707/893 centered states at $\varepsilon = 0.05$ and all 893 at $\varepsilon = 0.10$ admit a perturbation that pushes lateral offset beyond the ± 10 m envelope.

5.2.1 Pixel boxes and local-PCA regions

Effect of per-pixel independence. The mean-constrained L_∞ ball still permits arbitrary independent per-pixel deviations. To test how much of P3's CTE fragility requires those directions, we re-verify under the label-conditioned box and local-PCA regions, both with mean(x) = 0.5 and clipping enforced. Each local-PCA anchor carries its own neighbor set and PCA basis. The expanded run covers all 893 centered images for α=1σ and 2σ, and a deterministic 200-anchor subset for the more expensive α=3σ; the selection rule, region parameters, PCA dimension, and 300 s per-query timeout are identical across sample sizes.

Property	Region	Result	Failure axis / note
P1-global	R_1	Violated; three of four sub-queries SAT	CTE both; HE lower
P1-patterned	R_1	16/24 patterns proved	CTE in 8 patterns
P1a, P1b	R_3	UNSAT in 8D decoder-box; random 8D affine also UNSAT (dimensionality)	none in-region
P2 (ε=0.02)	R_1	0/893 robust on centered subset (69% wrong, 31% fragile)	Both
P3	R_2, R_4	R_2: 186/893; $R_4(1\sigma)$: 847/893; rand. 8D(1σ): 893/893	Dim.; CTE SATs
P4 base	R_1	SAT, CTE=0 (boundary)	wrong-sign
P4 sweep	R_3	SAT at $\lvert\text{CTE}\rvert \geq 3$	wrong-sign (1.7% emp.)
P6	R_1	CTE $[-24.3, 23.3]$	per-axis env.
P1/P4 (math)	R_6	SAT at all main calibrations; most stored witnesses rejected by both classifier audits	requires witness audit

Table 4 Summary of region-tagged TinyTaxiNet verification results. The Region column uses the tags defined in Section 3.1; detailed P3 status counts appear in Figure 3. The random-control entry is the matched 8D control at 1σ; at 2σ and 3σ the control also verifies more often than local PCA but is not witness-free.

Restricting the perturbations to local-PCA directions raises the verification rate substantially. The mean-constrained L_∞ ball verifies 186/893 anchors (20.8%), compared with 847/893 (94.8%) for local PCA at $\alpha = 1\sigma$ and 664/893 (74.4%) at $\alpha = 2\sigma$. This reproduces the contrast in the original 20-anchor sample, where the counts changed from 4/20 to 20/20 and 17/20. At 3σ, 64.0% of the deterministic 200-anchor subset is verified. The larger sample also reveals failures missed by the small sample: 43 anchors are SAT at 1σ and 225 at 2σ. Local PCA therefore restricts many failing directions while retaining some CTE-side counterexamples. At 1σ, every witness is CTE-side, with no HE violation among the 893 anchors; HE-side witnesses appear at the looser scales (8 of 893 at 2σ and 20 of 200 at 3σ).

The label-conditioned per-pixel box yields little improvement over the same-radius L_∞ ball (5/20 versus 4/20). Although its bounds come from same-state neighbors, the solver can still vary all 128 pixels independently. Local PCA removes those independent directions. This comparison identifies the region's structure as a factor in the verification-rate change; the next control tests the separate effects of dimension and orientation.

Matched-dimension control. To separate dimension from orientation, we replace the PCA basis with deterministic zero-mean orthonormal random 8D directions over three seeds. Each control retains the anchor's local-PCA per-direction scale σ_j, mean invariant, clipping, thresholds, and solver settings, so only orientation changes. The bases are orthonormal and zero-mean to 10^{-15}.

Random subspaces verify at least as often as local PCA at every scale: 893/893 at 1σ, 881/893 (median; range 877–886) at 2σ, and 169/200 (165–171) at 3σ, with no timeouts. The corresponding local-PCA counts are 847/893, 664/893, and 128/200. The improvement over the mean-constrained L_∞ ball can therefore be obtained through a low-dimensional restriction alone.

PCA's data-aligned directions are more effective at finding failures: they recover 43 versus 0 SAT anchors at 1σ, and 225 versus a median of 12 at 2σ. Thus, a higher verification rate can accompany lower coverage of data variation. Figure 4 shows this relationship: random 8D regions verify most often but capture little same-state variation, while local PCA captures more variation and exposes additional CTE-side witnesses.

Dimension sweep. We next test how the PCA/random difference changes with dimension. With α=1σ, we sweep $d \in \{2, 4, 8, 16, 32, 64\}$ on the 694 centered anchors with at least d+1 same-state neighbors at d=64 (Fig. 5). Both bases verify 100% at $d \leq 4$; at d=8, PCA verifies 94.4% and random verifies 100%. The gap widens at $d \gtrsim 16$: PCA verification falls to 68.4%, 38.8%, and 26.8% at d=16, 32, 64, while random remains at 100%, 99.8%, and 99.3%. The PCA runs have 18, 33, and 26 timeouts, respectively, which are not counted as verified.

The additional PCA directions expose mainly cross-track failures. On the 200-anchor candidate subset, we compare the 159 anchors with at least 65 same-state

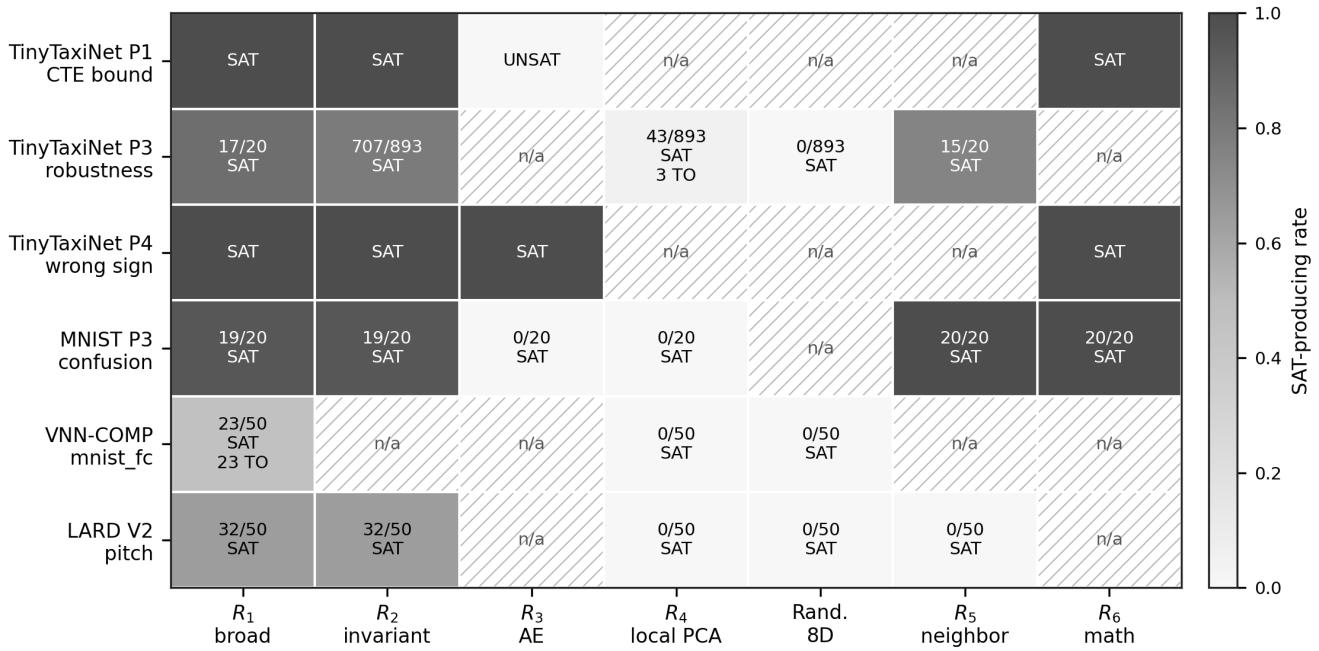


Fig. 2 Verdict changes across regions. Colors show the fraction of queries producing SAT, hatched cells were not tested, and "TO" denotes timeouts. Rows differ in sample size and timeout profile; interpret the rates using their printed denominators. TinyTaxiNet P3 uses all-893 counts for R_2, R_4, and the matched random 8D control, but the original 20-anchor counts for R_1 and R_5. The `mnist_fc` row shows the initial broad-ball run; Table 6 gives the counts after rerunning timeouts. Section 5.1 reports the P1 R_3 matched random-affine control; the compact-MNIST learned-region queries have no matched random-8D controls. The patterns distinguish removal by preprocessing or low-dimensional restriction from the persistent P4 wrong-sign behavior.

neighbors at every dimension. The number of anchors with a CTE counterexample rises from $0, 0, 9, 43, 95$ to 107 at d=$2, 4, 8, 16, 32, 64$, while HE-side counterexamples appear only at $d \geq 32$ (2 then 7 anchors). The data-aligned directions therefore reach the ± 10 m CTE envelope before the $\pm 90°$ HE envelope. The LARD replication (queries ~ 1.6 s each) gives a different pattern: both bases remove all broad witnesses at every $d \leq 64$, consistent with those witnesses lying outside the data-variance envelope at the matched scale.

Comparison with a subspace attack. We test whether a multi-restart corner-jump/PGD attack can recover the failures found by Marabou on the same local-PCA regions. The attack uses 30 restarts and 50 iterations per objective, takes 0.10 s per region, and accepts only valid-pixel witnesses. At d=8, it finds 0 witnesses against 33, 225, and 70 Marabou SAT verdicts at α=$1, 2, 3\sigma$, with 0 attack-only witnesses. Marabou also returns 622, 664, and 128 UNSAT certificates, respectively.

These comparisons use anchors with both attack and solver records. At α=1σ, the records cover 657 of the 893 anchors, including 33 of the 43 Marabou SAT anchors; the remaining 10 SAT anchors have no matching attack record. The 2σ and 3σ comparisons cover the full local-PCA runs. The attack's best objective falls short of the P3 bound by a median of 6.4 at d=8. Even a 500-restart stress run on a representative SAT anchor reaches only CTE $= -1.4$, compared with Marabou's feasible -10. In these tests, formal solving both recovers witnesses missed by the attack and proves bounds that attack failure alone cannot establish.

The local-PCA witnesses have mixed audit support: across the 14 CTE-side witnesses recovered at $\alpha = 2\sigma$ and 3σ on the 20-anchor sample, median classifier probability is 0.07 (real images: median 0.95), median nearest-real-image L_2 distance is 0.80, and 4/14 score above probability 0.9.

Sensitivity and output margins. To explain why most failures occur on CTE, we compare each output's sensitivity with its distance from the specified bound. On all 893 centered images, the exact local Jacobian $J(x) = \partial(\mathrm{CTE}, \mathrm{HE})/\partial x$ (finite-difference relative error below 10^{-11}) gives median norms $\|J_{\mathrm{HE}}\|_1 = 649$ vs. $\|J_{\mathrm{CTE}}\|_1 = 259$ and $\|J_{\mathrm{HE}}\|_\infty = 41.7$ vs. $\|J_{\mathrm{CTE}}\|_\infty = 8.5$. The network is therefore 2.5–4.9× more pixel-sensitive on HE than on CTE in these norms. The difference is explained in first-order terms by the margin between each prediction and its output bound: median centered prediction is $|y_0| = 1.68$ m vs. $|y_1| = 14.7°$, leaving median headroom of 8.32 m and 75.3°. The median first-order pixel radius that saturates each bound

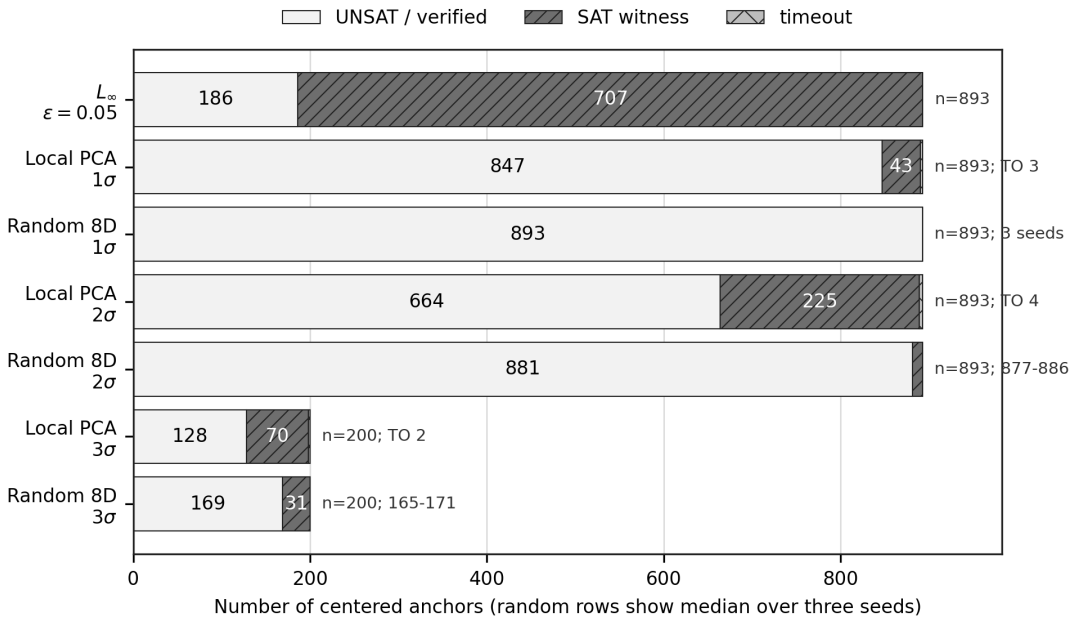


Fig. 3 Expanded P3 status counts generated from the released artifacts. The L_∞ row is the full centered mean+clip run at $\varepsilon = 0.05$; local PCA and matched random 8D controls at 1σ and 2σ cover all 893 centered images, while the 3σ rows use the deterministic 200-anchor subset. Random rows report the median count over three seeds with the verified-count range printed at right when nonzero. At 1σ, the random controls have zero SAT witnesses, while local PCA has 43 CTE-side SAT witnesses. Timeouts are reported separately and are not counted as verified.

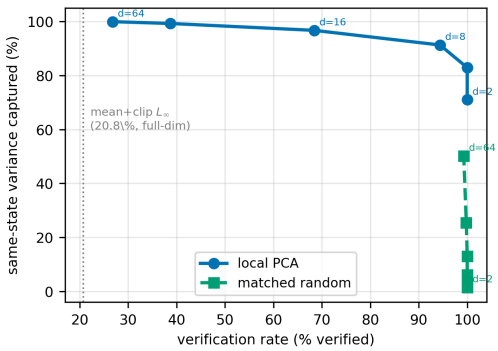


Fig. 4 Verification rate versus same-state variance captured by the subspace (TinyTaxiNet P3, α=1σ). Random subspaces verify more often but capture little data variation; local PCA captures more variation and therefore finds CTE-side witnesses as d increases. The full-dimensional L_∞ point has no comparable low-dimensional membership metric.

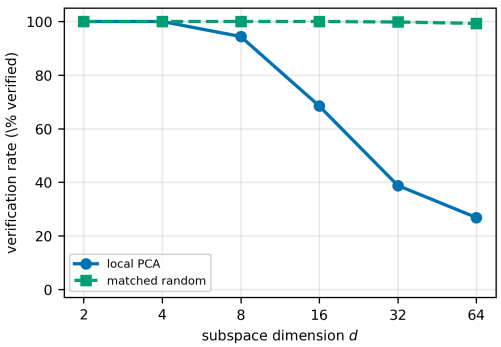


Fig. 5 P3 verification rate versus subspace dimension d at α=1σ (TinyTaxiNet, 694 anchors). Local PCA and the matched random subspace agree at small d but diverge at $d \gtrsim 16$: PCA verification falls as its data-aligned directions reach the CTE bound while the random subspace stays near 100%. Timeouts are reported separately and not counted as verified.

is $\varepsilon^{\text{sat}}_{\text{CTE}} = 0.0345$ vs. $\varepsilon^{\text{sat}}_{\text{HE}} = 0.121$, so the tested $\varepsilon \in \{0.02, 0.05, 0.10\}$ span the CTE radius but remain at or below the median HE radius.

Controlled postcondition sweep. Holding the region fixed (mean-constrained L_∞ ball, ε=0.05) and varying only the postcondition bound confirms the headroom explanation. The first-order violation fraction crosses 50% at a CTE bound of 15 m and an HE bound of 50°, close to the corresponding headroom predictions of 14.7 m and 47.2°. Setting the HE bound to 35.4°, which equalizes the median saturating radius with the CTE 10 m bound, removes the original CTE-dominated pattern: at ε=0.0345 the two axes violate at matched rates (50.2% CTE, 50.1% HE), and at ε=0.05 the exact Marabou run on the 20-image P3 sample changes from CTE 16/20 vs. HE 0/20 at the original bounds to CTE 16/20 vs. HE 15/20 under the equalized HE bound.

5.3 P4: directional failure across regions

P4 tests directional sign consistency on the left and right offset domains. Both C_{left} and C_{right} return **SAT** under the raw per-pixel box. The initial baseline witnesses lie at the programmatic boundary of the strict sign inequality (CTE = 0.0 m), establishing feasibility without characterizing failure severity. The empirical baseline shows that the same qualitative failure mode is already present on real dataset images: 92/4,128 left images (2.2%) and 50/4,099 right images (1.2%) produce wrong-sign CTE.

Mean-constrained and structured follow-ups: Enforcing mean(x) = 0.5 preserves the SAT verdict. The released threshold sweep shows that P4-left wrong-sign witnesses survive under R_2 at CTE ≤ -20 m; the same threshold remains SAT when R_2 is intersected with the tested math filters, including TV ≤ 2 and Fourier ≤ 0.62, both below the empirical centered-subset minima. The C_{right} base query is similarly SAT. Empirical wrong-sign magnitudes reach -3.51 m on the left-offset subset and $+3.28$ m on the right-offset subset (Section 6.1).

Empirical and regional evidence. Wrong-sign errors are concentrated near the centerline. The rate is 143/432 (33.1%) in the band $0.5 < |\text{CTE}| < 1$ m, 142/8,227 (1.7%) in the P4 offset subset ($|\text{CTE}| > 2$ m), and 41/5,490 (0.75%) beyond $|\text{CTE}| > 5$ m. The maximum wrong-sign magnitudes in these bands are 3.89 m, 3.51 m, and 2.28 m, respectively.

Within the offset subset, the rate falls from 6.3% at $|\text{CTE}| \in [2, 3)$ m to 0.5% at $[5, 8)$ m. Most incorrect signs also have small predicted magnitudes: 58% have |predicted CTE| < 1 m, and 12% have |predicted CTE| > 2 m. The wrong-sign images cluster more tightly than random offset images (mean pairwise L_2 0.97 versus 1.13), and each image's nearest real neighbor is on the same side. Together, these observations locate the failure near the sign boundary and give it direct dataset support. Wilson intervals treat the released 10,000 images as a sample from the simulator's image-generating process; the dataset rates themselves are exact counts.

The AE-composed sweep preserves this sign failure inside the subset-specific decoder-box regions, with witnesses up to -3.42 m and $+3.00$ m. Thus, P4 has both empirical support and witnesses under the learned regions. The much larger wrong-sign outputs admitted by the math-constrained regions require a separate witness-support assessment, reported in Section 6.3.

Contrast with Boeing TaxiNet. Pasareanu et al. [8] report *zero* blatant left/right confusion on the experimental Boeing TaxiNet (24-layer CNN, ELU activations): in their confusion matrix, no ground-truth "far left" CTE label is ever predicted "right" or vice versa.

5.4 Supporting diagnostics

P2 (local correctness). P2 distinguishes errors already present at the anchor from failures introduced by a perturbation. Under an L_∞ ball of radius ε=0.02 with a 1.0 m CTE band, it verifies 0/893 centered images under both R_1 and R_2. Relaxing the CTE band to 1.5 m leaves the count at 0/893. Of the centered images, 615/893 (68.9%) are already wrong at x_0, while 278/893 (31.1%) are nominally correct but fail somewhere inside the ball. The 20-image sample tracks the P3 verification rates but has a different P2 wrong/fragile split (9/20 versus 11/20), so the full-subset counts are used for this diagnostic.

P6 (box-envelope width). P6 resolves the absolute output envelope over C_{centered} by binary search (52 queries, tolerance 0.01). The certified bounds are CTE $\in [-24.33, 23.30]$ m and HE $\in [-90.10, 84.07]^\circ$ against empirical centered-prediction ranges $[-4.95, 4.86]$ m and $[-32.0, 33.3]^\circ$. These 4.86× and 2.67× over-approximation ratios measure the size of the box-abstraction headroom that P1d exploits.

5.5 Probe synthesis

The probes show why a verdict needs both a region description and supporting data. Enforcing a known preprocessing invariant can remove an invalid input combination, while restricting dimension can make a property provable even with random directions. Persistence across regions becomes stronger evidence of a model failure when the same behavior also occurs in the dataset, as in P4.

Using the workflow on a new witness. For a SAT witness recovered under a broad region R_1, rerun the property under tighter tagged regions and compare learned regions with matched random 8D controls at the same scale. Audit the recovered witnesses against the three dataset-support signals and retain the verdict, including timeouts, for each region. These records distinguish effects of preprocessing constraints, low-dimensional restriction, and behavior also observed in data. Poor audit scores identify candidates for further investigation; direct occurrence in the dataset or reproduction through a known renderer supplies stronger support. The controlled-oracle calibration tests this interpretation in a domain with known ground truth.

5.6 Transfer check on MNIST

We apply the same three-probe structure to a compact MNIST ReLU MLP ($784 \to 32 \to 16 \to 10$, 48 hidden ReLUs, test accuracy 96.94%) as a transfer check: one network, 20 anchors, three class-confusion properties, and the same three audit signals. The six region tags are recalibrated to MNIST: per-image or per-class pixel boxes, a class-conditioned total-mass band, class-specific ReLU autoencoder boxes, local PCA over same-class neighbors, nearest-neighbor same-class boxes, and per-class total-variation bounds. MNIST has no exact preprocessing equality comparable to TinyTaxiNet's mean invariant, so its R_2 is a statistical band. The learned regions are coarser than in TinyTaxiNet: class-AE reconstruction L_∞ median is 0.53–0.85 per class, and the top-8 local-PCA directions capture 75% of same-class-neighbor variance on average, versus 92% on TinyTaxiNet.

Table 5 reports the MNIST verdict rates. Broad pixel regions admit class-confusion witnesses, while the learned AE and local-PCA regions remain UNSAT over their encoded supports. No matched random-8D control is reported for this compact MNIST MLP, so the R_3/R_4 UNSAT columns are encoded-region verdicts rather than evidence that learned orientation drives the flip.

The class-confusion results show that the protocol can be applied to a second image model. The persistent-failure finding remains specific to TinyTaxiNet; the compact MNIST test instead motivates checking how witness-support measures behave in a different image domain.

The MNIST witness audit shows that the support signals must be recalibrated by domain. Real MNIST test digits have classifier probability 0.992, nearest-same-class L_2 median 4.77, and per-pair violation near 0.05. The per-image R_1/R_2 witnesses share the same 41 witnesses because the total-mass band is non-binding; they score low by classifier probability (mean 0.049) and high by per-pair violation (0.230), even though their nearest-real L_2 median is only 1.04. The R_5 label-conditioned witnesses have classifier probability 0.898 and per-pair violation 0.023 but nearest-real L_2 median 4.84, near the real baseline. The external `mnist_fc` run below adds the matched random-8D control for this broad-versus-low-dimensional pattern.

5.7 External instantiation: VNN-COMP mnist_fc

To test the workflow on an externally supplied classifier, we use the VNN-COMP `mnist_fc` network `mnist-net_256x2` (784−256−256−10 ReLU). We select 50 correctly classified held-out anchors (5 per class, ascending index) and fix train-only region calibration before solving. The property is targeted class confusion: $\text{logit}_t \geq \text{logit}_c$ for the anchor's highest non-true class t.

We compare a pixel ball R_1 at radius ε with local-PCA regions R_4 encoded as a composed ONNX network $z \mapsto x_0 + V_d^\top z \mapsto$ `mnist_fc`. Marabou 2.0 solves over the z-box $|z_j| \leq \alpha\sigma_j$, using d=8 and same-class training neighbors. We vary the broad-ball radius, $\varepsilon \in \{0.05, 0.03\}$, and the local-box scale, $\alpha \in \{1, 2, 3\}\sigma$.

Every bounded query enforces valid pixels. The composed network outputs the 784 reconstructed pixels, and the specification requires $0 \leq x_i \leq 1$; witnesses are scored without clipping. Before the experiment, we validate the Marabou output-index map (logits Y_0–Y_9, pixels Y_{10}–Y_{793}) by reproducing a confusion-only verdict from a single-output logits-only network.

Bounded local PCA at 1σ is UNSAT on all 50/50 held-out anchors and removes every resolved broad-region witness (Table 6). At ε=0.05, the initial broad-ball run finds class-confusion witnesses on 23 anchors, all of which become UNSAT under valid-pixel R_4 at 1σ (31–308 s, no local-PCA timeouts). A 10×-budget rerun (1200 s) of the 23 broad-ball timeouts resolves 2 SAT and 2 UNSAT cases, leaving 19 unresolved. Both additional witnesses are removed by local PCA and the matched random 8D control, bringing the resolved broad-witness count to 25. The remaining timeouts prevent a complete broad-ball classification.

The same removal occurs at ε=0.03: its 7 broad witnesses are a strict subset of the ε=0.05 set, and all 7/7 become UNSAT at 1σ. Increasing the local-box scale admits a counterexample: the sole 2σ SAT anchor is UNSAT at 1σ, and its witness passes the pixel-cube and confusion-replay checks.

The matched random 8D control is UNSAT on all 50 anchors at 1σ, 2σ, and 3σ (450/450 queries, no timeouts). It uses zero-mean orthonormal directions, the same per-anchor σ scale, and the valid-pixel composed-ONNX encoding, with revalidated indexing and seeds 0–2. Since it removes the same resolved broad witnesses, this experiment also attributes their removal to a bounded low-dimensional restriction. Class-AE R_3 queries are omitted here because the bounded-output encoding is computationally expensive for the solver.

5.8 External runway instantiation: LARD V2

To test the workflow on an independent source of runway images, we use LARD V2 Google Earth Studio imagery [34], independent of VerifyGAN/TinyTaxiNet in dataset, image generator, airports, and authorship. The

Region	P3 SAT	P1 SAT	P4 SAT
$R_1(x_0, \varepsilon)$ per-image L_∞ ball	2/20, 19/20, 20/20 at $\varepsilon = .01, .05, .10$	10/10	10/10
$R_2(x_0, \varepsilon)$ ball + total-mass band	2/20, 19/20, 20/20 at $\varepsilon = .01, .05, .10$	10/10	10/10
R_3 class-AE decoder-box composition	**0/20**	**0/10**	**0/10**
R_4 local PCA, $\alpha = 2\sigma$	**0/20**	**0/10**	**0/10**
R_5 label-conditioned per-pixel box	20/20	n/a	n/a
R_6 class box + TV at $\tau = p_{90}$	20/20	10/10	10/10

Table 5 MNIST verdict rates. P3 uses 20 anchors; P1 and P4 use one class-level or representative query per class. Regions R_3–R_6 are fixed by class or neighborhood and therefore have one P3 verdict independent of ε.

experiment uses 2000 training images and 700 held-out test images, resized to 24×48 grayscale and normalized to $[0, 1]$ without TinyTaxiNet-style mean-centering.

LARD provides public runway-corner and attitude annotations; our lateral-offset candidate is derived from the annotated runway-corner centroid, $\frac{1}{4}\sum_k x_k/\text{width} - 0.5$, and binned into train-set terciles. Target selection is performed on a validation split carved from the training subset: among seven annotation-derived targets, camera pitch is the only reliably learned target at this resolution (validation accuracy 0.820, held-out test accuracy 0.837 for the target-selection model), while runway-relative geometry targets such as lateral offset, slant distance, and glideslope remain near chance. The verification target is pitch-attitude classification, the coarse runway-image target with reliable signal at this resolution.

The verified model is a ReLU MLP $1152 \to 64 \to 32 \to 3$ trained from scratch on pitch terciles with ordinary supervised cross-entropy (test accuracy 83.0%). Training is separated from verification: the classifier is fit once on pitch labels, the manifest fixes 50 correctly classified held-out anchors before solving (class quotas 17/17/16), and the region constructors use training images only. The `.nnet` export reproduces the NumPy logits in Marabou to 5×10^{-9}. The property is targeted class confusion. The encoded regions are R_1 per-anchor L_∞ balls, R_2 the same ball with a class mean-brightness band, R_5 a same-class nearest-neighbor pixel box, and R_4 local PCA with and without valid-pixel constraints. All 750 Marabou queries complete with no timeouts (total recorded solver time 110.5 seconds; max per query 1.6 seconds).

The LARD comparison tests whether broad-region failures persist when the input variation is constrained by the dataset. Witnesses occur at the stress radius $\varepsilon = 0.4$; all anchors are UNSAT at $\varepsilon \leq 0.3$. Although the $\varepsilon = 0.4$ witnesses have valid pixel values, they differ substantially from held-out images. Their per-pair Lipschitz violation is 0.199 versus a real-image baseline of 0.010, nearest-real same-class L_2 distance is 11.8 versus 2.81, and total variation is 370 versus 96.

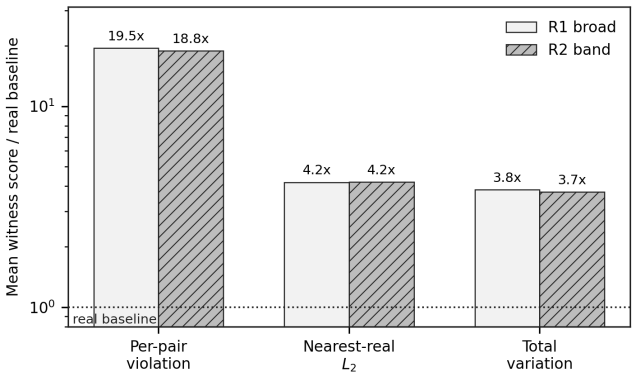


Fig. 6 LARD witness-audit ratios for the broad SAT witnesses. Values are means over recovered witnesses divided by the held-out real-image baseline. The R_1 and same-extent R_2 witnesses are valid pixel vectors, but they are about 19× above the real per-pair violation baseline, 4× farther from nearest same-class held-out images, and nearly 4× higher in total variation.

The same-pixel-extent R_2 control leaves all 32 witnesses SAT, showing that a brightness band alone does not account for their removal. A matched random 8D control removes every broad witness removed by local PCA. With zero-mean orthonormal directions, the same per-anchor σ scale, and seeds 0–2, all 900/900 queries are UNSAT at $1/2/3\sigma$, with and without valid-pixel constraints. These controls attribute removal to the low-dimensional restriction, while the audit measures how the witnesses differ from the data. The finding concerns pitch-attitude classification; the earlier target-selection experiment did not support verification of fine runway-centerline geometry at this resolution.

5.9 Convolutional front-end scalability probe

To test the workflow on a convolutional network, we trained a small ReLU CNN on the TinyTaxiNet regression task (`Conv→Conv→FC`, test CTE MAE 1.40 m), exported it to ONNX, and ran the P3 CTE-bound subqueries through the same region-tagged workflow on eight centered anchors. The broad pixel-ball queries produce 12 SAT and 2 UNSAT verdicts; two additional subqueries return non-SAT/UNSAT Marabou status codes recorded as solver errors by the exact parser.

Instantiation	Broad region	Tighter regions and controls	Purpose
VNN-COMP `mnist_fc`	R_1 pixel ball at $\varepsilon = 0.05$: 23 SAT / 4 UNSAT / 23 TO, then 25 / 6 / 19 after a 10× rerun of the timeouts	R_4 local PCA with valid pixels: 0/50 SAT at 1σ; 1/50 SAT at 2σ; matched random 8D: 0/50 SAT at $1/2/3\sigma$	Non-authored image classifier
LARD V2 pitch	R_1 pixel ball: 0/50 SAT at $\varepsilon \leq 0.3$ and 32/50 SAT at $\varepsilon = 0.4$; same-extent R_2 mean band also 32/50 SAT	R_5 nearest-neighbor box: 0/50 SAT; R_4 local PCA: 0/50, 0/50, 2/50 SAT; valid-pixel R_4: 0/50 SAT at $1/2/3\sigma$; random 8D: 900/900 UNSAT	External runway audit check

Table 6 External instantiation summary. Both checks use data and networks outside TinyTaxiNet/VerifyGAN and report broad-region witnesses removed by tighter encoded regions. Matched random-subspace controls are reported for all three instantiations (TinyTaxiNet P3, LARD, and `mnist_fc`); a random 8D subspace at matched scale removes the broad witnesses at least as completely as local PCA.

The local-PCA queries produce 14 UNSAT, 1 SAT, and 1 timeout at the 300 s budget. The run demonstrates application to a convolutional front end within the reported solver budgets. No matched random control was run for this CNN, and the conv-AE P1a/P1b queries time out at the reported budget.

5.10 Prior-Result Cross-Checks

The comparisons with prior TaxiNet-family results help place the findings in context. P3 at ε=0.02 verifies 892/893 anchors, qualitatively consistent with Julian et al.'s adaptive stress testing, which finds no failure at $\delta_{\max}$=0.02 under a $|d|$>10 m centerline-offset threshold [9]. For P4, the independently recomputed wrong-sign rate is 1.7% (142/8,227, 95% Wilson CI 1.47–2.03%). This differs from the zero blatant left/right confusion reported for the experimental Boeing TaxiNet [8]. The released empirical-baseline script reproduces the dataset calculations.

6 Witness Audits and Region Stress Tests

We audit representative P1 and P4 witnesses from the unconstrained and mean-constrained regions to assess their support in the dataset. The P3 results and per-image categories are reported in Section 5.2, and the P2 diagnostic is in Section 5.4.

The audit adds empirical evidence to each SAT verdict. A witness found only in a broad pixel region and scoring poorly on all support measures is a candidate abstraction artifact. Persistence under data-conditioned regions, together with direct occurrence in the data or reproduction through a known renderer, supplies stronger support. When the measures disagree, further data or a physical rendering model is needed to resolve the interpretation.

6.1 Pixel-box witnesses

The two unconstrained witnesses (P1a, P4 left) violate mean(x) = 0.5 (witness means 0.611 and 0.472) and are inconsistent with the released preprocessing invariant. Both failure modes survive the mean constraint with witnesses that satisfy the invariant: P1a at CTE=10.0, and P4-left as a wrong-sign SAT with achievable magnitude CTE ≤ -20 in the released threshold sweep. Across the P1a and P4 mean-constrained counterexamples, the nearest real images are at L_2 distances of 1.6–1.8, indicating that counterexamples combine pixels from different images in ways that do not occur in the dataset but are permitted by the per-pixel box abstraction.

Per-pixel bounds permit combinations absent from the training data: 0% of real images have more than half their pixels within 20% of the box width of either per-pixel bound, and the formal P6 bound is 4.86× wider in CTE than the empirical centered range. Mean-constrained SAT survival (Section 5.1, P1a/P1b/P4 all remain SAT under mean(x)=0.5) shows the failures are not artifacts of the missing mean invariant; the P4 wrong-sign mode is empirically present on 1.7% of offset images independent of any adversarial construction. A first attempt to tighten the region with linear statistical constraints (global PCA box on all 128 principal directions, 95% TV bound, 95% per-row mean bands) left P1a, P1b, P4 all SAT, which motivates the nonlinear autoencoder-composed region introduced next.

Autoencoder-composed verification. The first nonlinear tightening step is the AE decoder-box abstraction. Its construction (subset-specific autoencoder, decoder composed with TinyTaxiNet, latent bounding box, mean-sum auxiliary output) is described in Section A. For each property we apply the relevant subset autoencoder ($\text{AE}_{\text{centered}}$ for P1, AE_{left} and AE_{right} for P4) and verify the negated postcondition over the composed network with mean(x) = 0.5 enforced. Table 7 reports the pixel-box and AE-composed verdicts side by side.

Query	Pix+mean	AE+mean
P1a (CTE≥10)	SAT	**UNSAT**
P1b (CTE≤-10)	SAT	**UNSAT**
P4 left, base (CTE≤0)	SAT (boundary witness; magnitude floor ≤ -20)	SAT, 0.00
P4 left, sweep (CTE≤−3)	—	SAT, −3.42
P4 right, base (CTE≥0)	SAT, 0.00	SAT, 0.00
P4 right, sweep (CTE≥+3)	—	SAT, +3.00

Table 7 Pixel-box versus AE-composed verification, both with $\text{mean}(x) = 0.5$ enforced. Base rows use the original property negation; sweep rows tighten the wrong-sign magnitude threshold.

Autoencoder support audit. We assess the AE region through latent coverage, reconstruction error, and prediction changes. Train-only latent boxes cover $\geq$ 97.8% of held-out latents across the centered, left, right, and off-center subsets. Median reconstruction L_∞ is 0.118–0.160; median prediction shift is 0.66–0.77 m for CTE and 2.20°–3.79° for HE, with p95 shifts of 2.35–2.91 m and 7.59°–11.69°. The companion classifier accepts 86.7–97.3% of random latent-box decodes as "real," making it a permissive support screen. The classifier gives pixel-box witnesses logits of -8 to -32 and AE-composed witnesses logits of $+2.63$ to $+5.03$.

The AE decoder has linear output pixels, so we audit the decoded range of stored witnesses. The mean-constrained P4 AE witnesses remain inside the valid pixel cube: the strongest stored left witness has decoded range $[0.236, 0.842]$, and the stored right witnesses at $+2$ m and $+3$ m have ranges $[0.198, 0.767]$ and $[0.243, 0.761]$. Random latent-box controls also stay fully in $[0,1]^{128}$ for 100.0%, 99.1%, 98.6%, and 98.3% of centered, left, right, and off-center decodes, respectively. P1a/P1b have no AE witness to audit because those decoder-box queries are UNSAT.

Autoencoder stability analysis. Latent-box quantile sweeps, nearest-neighbor checks for ± 3 m P4 witnesses, and retraining at seeds 7 and 13 leave the AE-composed verdict pattern unchanged. The released artifact includes the scripts and outputs for these stability checks.

6.2 Learned-support results

The AE results support different interpretations for P1 and P4. For P1, the UNSAT verdicts prove the output bounds over the centered decoder-box region (Table 7). The reconstruction audit is needed to relate that region to the source images: reconstruction L_∞ reaches 0.27 and changes the predicted CTE and HE values.

For P4, wrong-sign behavior occurs in both the released images and the AE-composed queries. The ± 3 m AE witnesses have magnitudes consistent with the observed errors, although their nearest-real-image L_2 distance is ≥ 0.86; they are not copies of the empirical wrong-sign images.

Retraining with Marabou-compatible convolutional decoders preserves the P4 SAT result. P1a/P1b time out at 600 s, leaving those convolutional-decoder queries unresolved. The FC decoder's UNSAT results therefore remain specific to its encoded region and the tested budget. The per-seed autoencoder weights are released for reproducibility.

6.3 Math-encoded regions

The math-encoded region family R_6 intersects R_2 with linear-in-pixel constraints calibrated from empirical quantiles of the released dataset: total variation, Fourier high-frequency L_1, photometric basis residual, per-row mean, per-cell mean, and per-pair Lipschitz bounds. Across the tested calibrations:

1. *Failures persist under the main calibrations.* P1a/P1b, P4-left, and P4-right remain SAT under the main math-region calibrations: box+mean, TV $\leq$ 10, Fourier HF-L_1 $\leq$ 3.34, and photometric residual $\leq$ 0.184. An explicit threshold sweep on P4-left shows wrong-sign witnesses exist at CTE ≤ -20 m under every tested region, including TV ≤ 2 and Fourier $\leq$ 0.62 (both well below the empirical centered minimum).
2. *The stricter Fourier region excludes all centered dataset images.* Tightening Fourier to $\tau \leq 0.62$ (orthonormal, $\sim 10\times$ below the centered minimum HF-L_1) flips P1a/P1b to UNSAT, but the region admits 0/893 real centered images, so the UNSAT proves the queried output bound only over a region containing none of the observed centered images. The released artifact records the Fourier unit reconciliation.
3. *The witness audit rejects most math-region witnesses.* On the 91 math-region SAT witnesses for which the full pixel vector is stored, the original classifier scores 86/91 below probability 0.1 (94.5%; 95% Wilson CI 87.8–97.6%), and a disjoint-split retrained classifier scores 69/91 below 0.1. The median nearest-real-image L_2 distance is 1.41 (typical

real-to-real $\sim$ 0.8 on the centered subset), and the per-pair Lipschitz violation fraction is an order of magnitude above the centered-baseline rate of 0.009 on box-only witnesses and remains elevated on generic-math witnesses.

Classifier-calibration audit. The classifier is used as a support screen, and its calibration is checked three ways. The original classifier accepts 97.5%–98.6% of released real images across the centered, left, right, and off-center subsets, but these are training-set acceptance rates. A disjoint-split retraining accepts 98.9% of 1,000 never-seen real images and still rejects most math-region witnesses: median probability 2×10^{-4}, 69/91 below 0.1, 11/91 above 0.5. A fifth fake-family stress test with smooth low-frequency synthetic images shows the classifier's limits: the original classifier rejects 70.9% below 0.1 and admits 19.2% above 0.5, while the disjoint-split classifier rejects 43.9% below 0.1 and admits 36.4% above 0.5.

Under the original plausibility classifier, the real-centered baseline has median probability 0.991 and mean probability 0.934, with 41/45 images above 0.5. The 91 math-region SAT witnesses have median probability 0.000 and mean probability 0.035, with only 3/91 above 0.5; the disjoint-split retraining sensitivity is reported above.

The three audit signals measure different aspects of dataset support. Math-region witnesses have low classifier scores together with elevated nearest-image distance and per-pair violations; TinyTaxiNet R_5 label-box witnesses are much closer in nearest-image distance but are often rejected by the classifier; MNIST nearest-image distance is close to its real-digit baseline, while classifier and per-pair signals remain discriminating. These differences are why the audit retains all three measures for each witness.

Exact constraint definitions and Marabou slack-variable encodings are in Appendix A; script names, runway-layout variants, combined-constraint runs, and sensitivity reruns are released with the artifact.

Additional reruns. The released artifact includes math-region sweeps, runway-calibrated variants, P1-patterned and AE intersections, calibration-quantile checks, stratified witness audits, and failure-hardness curves.

7 Comparison with Prior Work

7.1 Region tightness and empirical support

Verification approaches differ in how they define the input set and how that set relates to the image source. Mitra et al. [3] distinguish verification of a perception network, a camera-plus-network composition, and a full closed loop. The Marabou pixel-box approach used here and by Wu et al. is sound and complete over the encoded region, which can include non-physical pixel combinations. VerifyGAN [7] instead restricts inputs to a learned generator range, leaving real images outside that range beyond the guarantee. Activation-pattern partitioning [6], local-PCA regions, and subset-specific autoencoders offer other ways to restrict the input set.

Simulation-based falsification complements these proofs. Fremont et al. [18] use VerifAI to search for failures in closed-loop simulation and redesign a TaxiNet-family taxiing network. Elboher et al. [4] study image-perturbation robustness on an Airbus runway-object classifier. Table 8 compares the input regions, guarantees, and limits of the directly related runway and TaxiNet-family studies.

7.2 Direct Runway and TaxiNet-Family Comparisons

Elboher et al. (DASC 2024). Their study verifies an Airbus runway-object classifier with Marabou under three image-perturbation families: pixel noise, brightness, contrast, and selected combinations. Their study is the closest runway-domain Marabou comparator: it verifies a larger industrial classifier and shows that robustness depends strongly on the perturbation family, with noise producing weaker robustness than brightness or contrast. Their main engineering contribution is query reduction: monotonicity across perturbation strengths and reuse of previous results reduce the number of verifier invocations by nearly 60%. The present work addresses a complementary layer of the same workflow: each verdict is tagged with the encoded region, learned/local/box/linear-filter regions are compared on a fixed property set, and recovered witnesses are audited for dataset support. The comparison therefore concerns two stages of verification: reducing repeated solver queries and assessing the support of the witnesses they produce.

Kadron et al. (VSTTE 2021). Their study verified TinyTaxiNet against $|y_0| < 10\,\mathrm{m}$ (which we adopt as one P1 sub-query) using Marabou under preconditions combining Prophecy-mined activation patterns with per-sequence pixel boxes; correctness thresholds (1.0 m CTE, 5° HE, which we adopt for P2) were evaluated on Boeing TaxiNet rather than TinyTaxiNet because Marabou could not handle the Boeing architecture's combination of convolutions and eLU activations. They report Marabou proofs for 33 of 111 time-interval sequences with ≥ 5 images (longest proved sequence: 17 images); our P1-patterned replication is single-image

Work	Input region	Network verified	Plausibility strategy	Main guarantee	Main limitation
Kadron et al. (VSTTE'21)	Prophecy-patterned per-sequence pixel boxes	TinyTaxiNet	Activation-pattern preconditions reduce loose pixel corners	33/111 time-interval sequences proved under $\|y_0\| < 10\,\text{m}$	Some partitions too broad to prove; soundness still relative to pixel box
Katz et al. (VerifyGAN, JAIS'22)	2-D state cell × 2-D GAN latent box	GAN generator ∘ TinyTaxiNet	Generator range defines the verifier inputs	Correctness band over state/latent cells	Real images outside generator support are outside the guarantee; verifies a composed network
Wu et al. (FMCAD'20)	L_∞ ball around individual pre-processed images	TinyTaxiNet	None; per-image local robustness	Per-image robustness with Marabou parallelization	No regional plausibility or global behavior claim
Elboher et al. (DASC'24)	Per-image perturbation regions: noise, brightness, contrast, and combinations	Airbus runway-object classifier	Perturbation family defined by image operation	Robustness assessment with nearly 60% fewer queries using monotonicity	No learned-support region or witness-support audit
This work	Six tagged regions: pixel boxes, mean invariant, AE decoder box, local PCA, label-conditioned box, linear diagnostic filters	TinyTaxiNet; subset decoder ∘ TinyTaxiNet for R_3	Region tags plus witness audit with classifier, nearest-neighbor, and Lipschitz signals	Verdict flips across regions; dimension/direction decomposition; P4 persists with empirical support	All guarantees are region-relative; AE and classifier are fitted support models

Table 8 Comparison with directly related runway and TaxiNet-family verification studies.

rather than per-sequence and verifies 16 of 24 observed centered patterns. Both reflect the same mechanism: tighter per-partition boxes eliminate non-physical corners that a global pixel box admits. Section 6.1 extends the same intuition nonlinearly via autoencoder composition.

Katz et al. (VerifyGAN, JAIS 2022). Their method moves verification out of pixel space by composing a conditional GAN with TinyTaxiNet and verifying over a 4-D box that pairs a 2-D state cell (CTE, HE) with the 2-D latent L_∞ ball, with the state space partitioned into 16,384 cells and the latent truncated to $[-0.8, 0.8]^2$. Every input Marabou considers is generated by the GAN, so the search is restricted to the generator range. Soundness is relative to the GAN-generated image set, not the underlying real-image distribution. Their pipeline verifies a different network (GAN∘TinyTaxiNet) over a different input space (the 4-D state-plus-latent box vs. our 128-D pixel-box and AE-composed regions), so direct numerical comparison is not possible. Our AE-composed region (Section 6.1) is a ReLU-only learned-support analogue implemented by decoder composition, with reconstruction L_∞ up to 0.27.

Wu et al. (FMCAD 2020). Their study examined local robustness queries on TinyTaxiNet using Marabou v1.x with a focus on parallelization (Split-and-Conquer); the input region is a per-image L_∞ ball and the postcondition is local robustness, with no plausibility step. Our P3 setup is the closest analogue and is complementary: we use Marabou 2.0 (queries solve without explicit Split-and-Conquer scheduling on this small network), evaluate at three dataset-calibrated ε values rather than one, and report both 20-image diagnostics and full centered-subset rates (n=893).

7.3 Broader Verification Landscape

This work uses deterministic verification at the perception level. Pixel-space verifiers [19,20,21,22] take the region as a user-supplied input. Closed-loop NN-controlled-system verifiers [23,24,25,26] work in state

space and assume a network with state inputs. Randomized smoothing [15], semantic-perturbation layers [16], and spatial-robustness studies [1] address related choices of input variation. Our comparison focuses on how the encoded region changes a fixed property verdict and how the resulting witnesses relate to observed images.

State-to-image verification. The most directly comparable predecessor is Yang and Rinard's Tiler [27], which verifies a camera-centerline regression network on a synthetic ray-traced road scene by formalizing the feasible input space as the image of an explicit observation process $g: \mathcal{S} \to \mathcal{P}(\mathcal{X})$ from a low-dimensional state space (camera offset δ, viewing angle θ) to 32×32 grayscale images, then tiling (δ, θ) and verifying per-tile robustness via MILP.

The setting is close to ours: camera-derived perception of a road centerline and regression of an offset and angle. The main differences are dataset source (idealized ray-tracing vs. X-Plane simulator), network class (ConvNet vs. ReLU MLP), and verifier (MILP-per-tile vs. Marabou with multiple explicit region constructions). Tiler defines one observation-process-induced region and decomposes it over a (δ, θ) grid; our protocol treats the region itself as the experimental variable and runs the same property set under six alternatives (R_1–R_6). Tiler's state-conditioned image set is the kind of region our R_3 (autoencoder decoder-box) and R_4 (local-PCA) regions approximate when no ray-traced observation model is available.

Geometric-perturbation verification. Estornell et al. [2] address the same input-support gap from the verification side: they construct a boundable state-to-image mapping using per-entity geometric transformations (translation, rotation, scaling, plus brightness/contrast modeled as an affine pixel-intensity map), propagate pixel-wise bounds through the controller using DeepG and CROWN-style linear-relaxation perturbation analysis (LiRPA), and produce closed-loop outer-approximations of reachable sets on CartPole, Pendulum, and Acrobot Gym environments with explicit soundness guarantees. Their guarantees rely on entity-decomposable scenes and an analytic state-to-observation model available at verification time. We work in the regime where only a dataset and statistical region models are available. In that setting, each verdict must be tagged with the region construction that produced it, and each recovered witness must be interpreted together with its audit signals.

Verification and validation of abstractions. Peper, Miao, Mitra, and Ruchkin [28] propose a unified verification-and-validation pipeline for vision-based autonomy in which an Interval-MDP abstraction is first model-checked at confidence $1-\beta$ and then statistically re-validated on a held-out dataset D_{val} under environment shift, yielding nested probabilistic guarantees on the abstraction's continued validity. Their validation target is the IMDP abstraction, a model used as input to verification; our audit target is the individual solver-returned witness. Both approaches treat empirical checking of verification artifacts as part of the verification workflow, but they operate at different levels: abstraction-level statistical validation in their case, witness-level audit in ours.

Position of this work. The input-support gap for pixel-space verification is well recognized, and constructive sound regions exist where a parametric observation, generator, or latent model is available (Tiler, Estornell et al., VerifyGAN, NNLander-VeriF [17], SEVIN [29]). SEVIN learns a VAE latent space and verifies image-controller properties over annotated convex polytopes; it is a constructive single-region method, while this work compares multiple region constructions on a fixed property set. Geng et al. [14] make a related argument that "data as specification is too tight" as motivation for Neural Activation Patterns, a delta-relaxed per-class subset-pattern construction distinct from our P1-patterned region, which uses full per-image Prophecy-style patterns [6,13]. Auditing solver-produced counterexamples is also a recognized concern: Song et al. [30] construct *unrestricted adversarial examples* entirely from a generative model's latent space, and Barredo-Arrieta and Del Ser [31] audit classifiers by generating *plausible counterfactuals* through a GAN under a multi-objective trade-off between model confusion and human plausibility.

This work combines region comparison and witness auditing in a reusable Marabou protocol. The same property set is evaluated under six explicit constructions, and each recovered witness is scored by one learned and two non-learned dataset-support measures. The matched random controls make it possible to distinguish the effect of a small input dimension from the effect of learned directions. The TinyTaxiNet analysis develops that comparison in detail, while the MNIST, external `mnist_fc`, LARD, and controlled-oracle runs test its use beyond the original simulator setting.

8 Reproducibility and Runtime

8.1 Reusable module and instantiations

The `region_audit.py` module implements the six region constructors and three witness-audit signals for flattened image-input networks verified through the `maraboupy` API. The module is parameterized by image

grid shape and calibration data. Its interface has four parts:

- **Region constructors.** `r1_subset_box` ($R_1(S)$), `r1_linf_ball` ($R_1(x_0, \varepsilon)$), two R_3 decoder constructors for in-query ReLU decoders or pre-composed decoder networks, `r4_local_pca` (R_4), and `r5_neighbor_box` (R_5). Each returns a tagged `Region` whose `attach(net, ivars)` adds its bounds and constraints to a Marabou network.
- **Intersectable constraints.** `mass_equality` and `mass_band` (the R_2 layer, exact invariant or statistical band), and the R_6 family: `tv_bound`, `fourier_hf_bound`, `photometric_bound`, `per_pair_bound`, `per_row_mean_band`, all linear-in-pixel using the slack-variable encoding of Appendix A. `intersect(region, ...)` composes any of them with any region under a new tag.
- **Tagged queries.** `run_query(make_net, region, postcondition)` builds a fresh network, attaches the region, applies the negated property (helpers: `output_at_least`, `output_at_most`, `class_confusion`), solves, and returns the verdict together with the region tag, its calibration parameters, and the witness vector when SAT. Region tagging is part of query execution and result serialization.
- **Witness audit.** `WitnessAudit` calibrates the per-pair Lipschitz thresholds from reference images and scores any witness vector with the three signals; `baseline()` reports the same signals on real images, so witness scores are always read against a same-domain reference. `load_relu_classifier` adapts the small ReLU plausibility classifiers used in both case studies; any callable returning a probability can be substituted.

An instantiation supplies dataset loading, subset predicates, neighbor rules, calibration data, and the network reader. The two released adapters, `taxinet_instantiation.py` and `mnist_instantiation.py`, also serve as smoke tests: 8/8 TinyTaxiNet checks and 7/7 MNIST checks reproduce representative recorded verdicts through the module, including the in-Marabou class-AE decoder composition. The per-experiment scripts remain the source artifacts for the paper's numerical results; the module packages the same constructions for reuse.

8.2 Artifact contents

The code and recorded results are available at https://github.com/Jireh1226/STTT-Region-Verification, commit `99ebb0007a31`. The repository's `RESULTS_AND_CODE.md` index links the reported findings to their scripts and saved outputs. Large model artifacts (`.npz`, `.keras`, `.onnx`, `.h5`) are regenerated by released scripts. The TinyTaxiNet weights and `SK_DownsampledGANFocusAreaData.h5` dataset are obtained from the upstream Stanford Intelligent Systems Laboratory `sisl/VerifyGAN` repository and are not redistributed. The environment is Python 3.11.14 with `maraboupy` 2.0.0, `numpy`, `h5py`, `tensorflow`, `tf2onnx`, and `matplotlib`; experiments were run on laptop-class macOS CPUs, with no GPU used for the LARD run. Marabou uses per-query timeouts only, with `normalize=False` for the `.nnet` reader.

The `Makefile` groups reproduction commands by dependency requirements. `make reproduce-empirical` recomputes TinyTaxiNet empirical summaries, Jacobian/headroom diagnostics, the controlled post-condition sweep, and classifier-support audits using only `numpy` and `h5py`. `make reproduce-solver` adds `maraboupy` for representative verdicts and exact-solver checks. `make reproduce-learned` runs the MNIST/TensorFlow components. `make reproduce-headlines` runs the first two tiers, providing the empirical and solver reproductions without TensorFlow.

The baseline query set is exported as 118 `.vnnlib` files in standard SMTLIB2 format with an `instances.csv` model map. Mean-constrained, local-PCA, AE-composed, math-region, classifier, threshold-sweep, controlled-oracle, MNIST, external `mnist_fc`, and LARD follow-ups are released as Python scripts plus matching JSON outputs; the artifact manifest maps each experiment to its script and output files.

8.3 Runtime

All 474 baseline queries complete within the 1-hour timeout: P1-patterned (96), P2 (160), and P3 (160) resolve at ∼10–20 ms per query (totals under 3 s); P4 is sub-second; P6's 52 binary-search queries take 70 s total. Peak memory is 80–108 MB.

The expanded local-PCA P3 run over all 893 centered images takes 1742 s at 1σ and 2760 s at 2σ with a 300 s per-sub-query timeout; the deterministic 200-anchor 3σ run takes 1096 s. The full-centered-subset reruns of P2 and P3 mean+clip complete in 183 s total wall-clock time across all ∼ 8,000 sub-queries. The follow-up math-region, sensitivity-rerun, stratified-audit, and threshold-sweep experiments complete in tens of seconds to a few minutes each; the most expensive single experiment is the conv-AE P1a/P1b query, which times out at the chosen 600 s budget on both centered sub-queries.

9 Calibration Checks

Two calibration checks test the solver interface and the interpretation of witnesses. First, we solve a deterministic 20-instance slice of the VNN-COMP `mnist_fc` benchmark using the same Marabou 2.0 binary and VNN-LIB handling as the main experiments. All 12 instances completed within the budget match the published SAT/UNSAT labels, with zero disagreements.

Second, a renderer-controlled lane-image domain supplies known ground truth for three constructed ReLU models: a correct model, an injected true-failure model, and an artifact-trigger model. Broad boxes are SAT for all three. The manifold-constrained local-PCA region removes the artifact failures and preserves the injected true failure; the nearest re-rendered state reproduces wrong-sign behavior only for the injected true-failure model. Individual distance and classifier-like scores do not separate all true and artifact cases. Combining region survival with reproduction through the renderer resolves those cases, supporting the use of multiple kinds of evidence in the witness audit.

10 Discussion and Validity Scope

The interpretation of a verdict depends on both the encoded input region and its relation to the application. We first describe the scope of the encoded guarantees, then the assumptions needed to relate them to deployed systems.

10.1 Scope of the Encoded Guarantees

1. **Binary-search tolerance.** P6 binary-search bounds are accurate to ± 0.01, not the exact extrema. The reported envelope is therefore an axis-aligned outer envelope to within this tolerance.
2. **Latent decoder-box restriction.** The AE-composed proof applies to the learned decoder over its training-derived latent box with $\text{mean}(x) = 0.5$ enforced (Section 6.1). Images outside this range are outside the guarantee. Reconstruction L_∞ reaches 0.27 and can change CTE/HE predictions, so the proof concerns reconstructed images. The support audit measures those changes; the matched random control in Section 5.1 tests whether the decoder is needed to obtain the P1a/P1b verdict change.
3. **Plausibility classifier scope and permissiveness.** The classifier is trained to discriminate real images from four synthetic fake types: per-pixel-box corners, cross-image blends, uniform noise, and high-frequency patterns. The original classifier's real-image acceptance is a training-set rate; the disjoint-split retrained classifier and the fifth fake-family stress test in Section 6.3 report the held-out and out-of-family behavior. The classifier also accepts most random AE latent-box decodes as "real" (Section 6.1). We therefore use classifier score as one dataset-support proxy, alongside nearest-neighbor distance and per-pair Lipschitz violation fraction.
4. **Multi-seed scope.** Multi-seed stability was tested over three autoencoder training seeds (42, 7, 13). Architecture choice, latent dimension, train/validation split, classifier seed, and classifier architecture are fixed. The reported stability covers SAT/UNSAT agreement and within-0.42 m witness magnitudes across these three seeds.
5. **Local-PCA support-model scope.** The local-PCA region uses $d = 8$, radius α, and same-state-neighbor bins as modeling parameters. The top-8 directions capture about 92% of same-state-neighbor deviation variance, leaving 5–11% outside the verified subspace depending on the anchor. These verdicts therefore characterize the encoded local linear region, not the full same-state image set.
6. **Matched-dimension control scope.** The comparisons cover three random seeds for TinyTaxiNet P3, VNN-COMP `mnist_fc`, and LARD, together with the P1a/P1b affine control. They support the dimension/direction interpretation within the tested networks, properties, and scales. Additional networks, subspace dimensions, and larger solver budgets at high d remain to be studied.
7. **Math-region filter scope.** The TV, Fourier, photometric, and per-pair-Lipschitz constraints are linear-in-pixel filters calibrated on the centered runway subset. They restrict selected image statistics; the dataset support of feasible witnesses is assessed separately by the audits in Section 6.3.

10.2 External Validity to Deployed Systems

1. **Hyperrectangular over-approximation.** Per-pixel boxes in R_1–R_2 admit non-physical pixel combinations. SAT counterexamples in R_1 need not be supported by the runway-image distribution; this is the core motivation for the AE-composed region R_3. Activation-pattern partitioning (P1-patterned) mitigates this for centered images but does not remove the region-relative scope.
2. **Preprocessing-invariant modeling.** The P1/P4 pixel boxes use dataset-derived bounds within $[0, 1]$. P3 mean+clip and local-PCA queries explicitly enforce $0 \le x_i \le 1$ (Section 5.2), while AE decoder queries do not; their stored witnesses and random decodes are range-audited in Section 6.1. These

constraints bound preprocessed inputs without encoding the operation $x \mapsto \mathrm{clip}(x, 0, 1)$ itself. The two agree only where clipping is a no-op. The mean equality is therefore a partial preprocessing model.

3. **Perturbation-space mapping.** P2/P3 perturb the already preprocessed network inputs. Their radii are calibrated against local dataset statistics (Section 3.3); they do not quantify raw-camera changes or weather conditions. Extending the verdicts to the full camera-to-output pipeline requires a model of that pipeline.
4. **Simulation dataset scope.** The 10,000 VerifyGAN images are simulator-generated (X-Plane) and represent one runway/lighting configuration. Kadron et al. used 51,462 labeled images. Our regional empirical baselines (e.g., 1.7% wrong-sign rate) are therefore in-distribution claims about the released VerifyGAN dataset, not about a particular real-world deployment.
5. **Local-PCA variation model.** The same-state-neighbor PCA region captures image variation among neighbors with similar (CTE, HE) labels. It cannot witness failures that arise from *between-state* distribution shift, such as states the data rarely visits. The local-PCA verdicts answer robustness questions within the encoded same-state subspace.
6. **Target network scope.** TinyTaxiNet is the main case-study network; Boeing TaxiNet is not formally verified or empirically evaluated in this work.
7. **Scale and architecture scope.** The verified networks in the main study are ReLU MLPs or ReLU decoder compositions of modest size. Section 5.9 reports a small CNN probe through ONNX; it demonstrates applicability of the workflow but also shows the solver-time boundary on a convolutional front end. The conv-AE P1a/P1b queries and the 28×28 Fourier encoding time out under the budgets reported here. Scalability to deployed-size convolutional or transformer perception stacks is not demonstrated.

10.3 Scope of Interpretation

The results characterize TinyTaxiNet as an open-loop perception model. The verification model excludes the controller, temporal filtering, and any downstream use of the HE channel, so the findings are diagnostic rather than closed-loop safety guarantees for an autonomous taxiing system. In deployment, perception NNs of this class are typically wrapped in run-time assurance architectures that monitor for and recover from perception faults (Cofer et al. [32] apply this pattern to the same Boeing TaxiNet that motivated TinyTaxiNet), and proposed certification frameworks for safety-critical NNs explicitly contemplate bounding the NN input/operational domain (ODD), which relates to the input-region encoding choices this paper diagnoses [33].

The region comparisons in Section 5.5 could inform run-time assurance or ODD-bounding analyses. Constructing and validating an online monitor would require a separate system-level study.

Semantic reachability and operational relevance. P4 has direct support within the released simulator dataset. Extending that finding to other runways, lighting conditions, or controllers requires evidence from those settings. Many wrong-sign cases are near the CTE decision boundary, so closed-loop rollouts are needed to determine whether the small sign errors diminish or accumulate under control. The controlled-oracle calibration (Section 9) distinguishes injected model failures from abstraction artifacts in a small rendered domain with known ground truth. TinyTaxiNet has no corresponding camera or rendering model; its region-survival results therefore remain evidence about the tested regions and dataset.

10.4 Outlook

Further work can connect the protocol to models of the full perception pipeline by composing preprocessing into the verified network, testing larger convolutional or transformer models through ONNX, and evaluating region comparisons within run-time assurance monitors. Where camera geometry, runway layout, and lighting parameters are available, analytic or hybrid state-to-image models could provide stronger support than dataset-fitted AE/PCA regions. Additional adapters for `region_audit.py` and a persistent DOI archive would make the artifact easier to reuse and cite.

11 Conclusion

This study provides a reusable Marabou protocol for interpreting verification results in relation to their input regions. It records the region and parameters for each verdict and audits recovered witnesses with classifier, nearest-neighbor, and per-pair Lipschitz measures.

The TinyTaxiNet experiments distinguish three reasons for a verdict change. A missing preprocessing invariant accounts for the P1 heading-side witness. A low-dimensional restriction accounts for the P1 decoder-box proofs and much of the P3 verification-rate increase, while data-aligned PCA directions expose failures that random directions miss. P4 persists across the tested restrictions and occurs directly in the released

simulator images. These findings show why the encoded region and empirical support are needed to interpret a proof or counterexample.

The MNIST, VNN-COMP `mnist_fc`, and LARD experiments demonstrate use of the protocol beyond TinyTaxiNet. Matched random controls on the two external runs support the same low-dimensional interpretation of witness removal. The released `region_audit.py` module and experiment records make these comparisons available for reuse: a practitioner can identify which region produced a verdict, which restrictions changed it, and how the recovered witnesses relate to the available data.

Acknowledgments

This study used the TinyTaxiNet model and Verify-GAN dataset from the Stanford Intelligent Systems Laboratory (SISL). Marabou 2.0 was used via the `maraboupy` Python API.

Appendix

A Region Construction Reference

This appendix gives the formal definitions, Marabou encodings, and calibration details for the six region tags R_1–R_6 of Section 3.1.

Raw per-pixel dataset box (R_1). For a subset S and pixel coordinate $j \in \{1, \dots, 128\}$, the raw dataset box uses

$$\ell_j = \min_{i \in S} x_i^{(j)}, \qquad u_j = \max_{i \in S} x_i^{(j)}.$$

Marabou encodes this region with 128 independent input lower and upper bounds, $\ell_j \le x_j \le u_j$. No cross-pixel constraint is imposed, so SAT witnesses may combine individually observed pixel values in joint configurations that no image in S contains.

Box plus mean invariant (R_2). The mean-constrained box starts from R_1 and adds the preprocessing invariant

$$\sum_{j=1}^{128} x_j = 64,$$

equivalent to $\mathrm{mean}(x) = 0.5$ on the flattened 8×16 image. In Marabou this is one linear equality over the 128 input variables.

Subset-specific autoencoder (R_3). One ReLU-only autoencoder is trained per input subset: $\mathrm{AE}_{\text{centered}}$ on 893 $|\mathrm{CTE}| < 1\,\mathrm{m}$ images, $\mathrm{AE}_{\text{left}}$ on 4,128 $\mathrm{CTE} > 2\,\mathrm{m}$ images, $\mathrm{AE}_{\text{right}}$ on 4,099 $\mathrm{CTE} < -2\,\mathrm{m}$ images, and $\mathrm{AE}_{\text{off-center}}$ on 9,107 $|\mathrm{CTE}| > 1\,\mathrm{m}$ images. The architecture is $128 \to 64 \to 16 \to 8 \to 16 \to 64 \to 128$, with ReLU hidden layers and a linear bottleneck, and training minimizes MSE reconstruction loss. On the training subset, the mean L_∞ reconstruction error is 0.12–0.15, with a maximum of 0.27. Let $D_S : \mathbb{R}^8 \to \mathbb{R}^{128}$ denote the trained decoder for subset S, and let $[\ell_k^z, u_k^z]$ be the coordinate-wise latent box over training encodings. The verified AE region is

$$R_3(S) = \{x = D_S(z) : \ell_k^z \le z_k \le u_k^z,\ k = 1, \dots, 8, \ \textstyle\sum_{j=1}^{128} x_j = 64\}.$$

For verification, the decoder ($8 \to 16 \to 64 \to 128$ linear) is composed with TinyTaxiNet into a single ONNX network mapping latent z through decoded pixels x to (CTE, HE), with an auxiliary output equal to the decoded pixel sum. The preprocessing invariant is enforced by constraining that auxiliary output to 64. The decoded pixels are not constrained to $[0, 1]$ in this composition.

Per-image local PCA (R_4). For each query image x_0 used in P3, the input region is constructed from same-state neighbors. We collect all images with $|\Delta\mathrm{CTE}| \le 0.5\,\mathrm{m}$ and $|\Delta\mathrm{HE}| \le 5°$ (between 42 and 89 neighbors per query, excluding x_0), form the deviation matrix $D \in \mathbb{R}^{m \times 128}$ with row i equal to $x_i - x_0$, compute the SVD $D = U\Sigma V^\top$, take the top $d = 8$ rows of $V^\top$ as principal directions V_d, and let σ_j be the standard deviation of the projection of D along direction j. For scale α, the verified region is

$$R_4(x_0, \alpha) = \{x = x_0 + V_d^\top z : |z_j| \le \alpha\sigma_j,\ j = 1, \dots, d, \ 0 \le x_i \le 1,\ \textstyle\sum_i x_i = 64\}.$$

Results are reported at $\alpha \in \{1, 2, 3\}$. The region is encoded in Marabou by introducing 8 auxiliary variables $z_1, \dots, z_8$ with bounds $\pm\alpha\sigma_j$ and 128 sparse linear equalities tying decoded pixels to the latent.

Label-conditioned per-pixel box (R_5). For a P3 anchor image x_{i_0} with label $(\mathrm{CTE}_{i_0}, \mathrm{HE}_{i_0})$, define the same-state neighbor set

$$N(x_{i_0}) = \{i \ne i_0 : |\mathrm{CTE}_i - \mathrm{CTE}_{i_0}| \le 0.5\,\mathrm{m}, \ |\mathrm{HE}_i - \mathrm{HE}_{i_0}| \le 5°\}.$$

For each pixel j, the encoded bounds are $\min_{i \in N(x_{i_0})} x_i^{(j)} \le x_j \le \max_{i \in N(x_{i_0})} x_i^{(j)}$, intersected with $0 \le x_j \le 1$ and the mean equality above. This uses the same Marabou encoding as R_1 and R_2, with bounds calibrated to the anchor's label neighborhood.

Math-encoded constraints (R_6). The math-encoded family starts from R_2 (box plus $\sum_j x_j = 64$) and adds one or more linear-in-pixel constraints, calibrated from empirical quantiles of the centered subset. Let E be the horizontal/vertical adjacency set on the 8×16 grid ($|E| = 232$). The total-variation region uses

$$\mathrm{TV}(x) = \sum_{(a,b) \in E} |x_a - x_b| \le \tau_{\mathrm{TV}}.$$

The Fourier region uses the orthonormal two-dimensional DFT

$$c_{k\ell}(x) = \frac{1}{\sqrt{128}} \sum_{r=0}^{7} \sum_{s=0}^{15} x_{rs} \exp\!\left(-2\pi i\left(\frac{kr}{8} + \frac{\ell s}{16}\right)\right)$$

and bounds high-frequency mass over $H = \{(k, \ell) : k \ge 4 \text{ or } \ell \ge 8\}$:

$$\sum_{(k,\ell) \in H} \left(|\Re c_{k\ell}(x)| + |\Im c_{k\ell}(x)|\right) \le \tau_F.$$

The photometric-basis region introduces two free variables (α, β) and enforces $\|x - \alpha b_1 - \beta b_2\|_\infty \le \varepsilon_P$, where $b_1 = \mathbf{1}/\sqrt{128}$ and b_2

is the unit-norm horizontal-gradient template. The runway-layout subfamily uses row, cell, and pairwise constraints calibrated from empirical centered-subset quantiles: $\ell_r \le \frac{1}{16}\sum_{s=0}^{15} x_{rs} \le u_r$ for each row r; $\ell_C \le \frac{1}{|C|}\sum_{(r,s)\in C} x_{rs} \le u_C$ for each 2×2 or 4×4 cell C; and $|x_a - x_b| \le \tau_{ab}$ for each adjacent pair $(a,b) \in E$. All absolute values are encoded with Marabou slack variables: for a linear expression $q(x)$, introduce $d \ge 0$ and impose $q(x) \le d$ and $-q(x) \le d$, then constrain the appropriate sum of d variables. The released scripts include separate TV, Fourier, and photometric-basis variants. Calibration values and witness audits are in Section 6.3.

Soundness scope. Tags R_1, R_2, and R_5 are exact box/equality regions. R_3 and R_4 are learned or local support models fit from the released preprocessed dataset; SAT/UNSAT verdicts under them are exact only over the encoded latent box or affine subspace, not the real-image distribution. R_6 is an explicit linear filter family; verdicts under it are exact over the linear-constraint-feasible set, and dataset support is assessed separately by the witness audit (Section 6.3).

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